Inventory product management
Ideation Phase
Brainstorm & Idea Prioritization

Date	09/03/2026
Team ID	SWTID-2026-4031
Project Name	Product inventory management system
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template

Brainstorming is used to generate creative ideas and identify the most suitable solution for a real-world problem. During the ideation phase, team members collaborated to discuss common issues faced in managing product inventories and explored multiple web-based solutions using the MERN stack. Ideas were evaluated based on feasibility, scalability, security, and learning outcomes.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

The team conducted a discussion session to identify common problems that could be solved using full-stack development. Several ideas were proposed such as:

- E-commerce product tracker
- Expense manager
- Task management system
- Cryptocurrency dashboard
- Inventory management system

After evaluating complexity, real-world usefulness, and alignment with MERN stack learning objectives, the team finalized the following problem statement:

Problem Statement:
Small businesses and individual users lack a simple, secure, and centralized system to manage their product inventory. Many existing solutions are complex or require paid subscriptions. Therefore, the team decided to build a **JWT-based Product Inventory Management System** that allows users to securely manage their own products.

Step-2: Brainstorm, Idea Listing and Grouping

During brainstorming, all team members contributed ideas without restrictions. The ideas were then grouped into categories:

Authentication & Security Ideas

- JWT-based login system
- User-specific product access
- Secure password hashing
- Role-based access (future scope)

Core Inventory Features

- Add product
- Update product details
- Delete product
- View all products
- Search products by name

User Experience Ideas

- Simple dashboard UI
- Responsive design
- Clean product cards layout
- Fast loading using React

Technical Ideas

- MERN stacK architecture
- REST API development
- MongoDB database design
- Middleware for route protection

After grouping, the team focused on ideas that were realistic within the project timeline.

Step-3: Idea Prioritization

The team prioritized ideas based on:

- Feasibility within deadline
- Learning value
- Practical real-world use

- *Technical complexity*

High Priority (Implemented Features)

- *JWT Authentication (Register/Login)*
- *Private product CRUD operations*
- *MongoDB database integration*
- *Secure API routes*
- *Search functionality*

Medium Priority (Enhancements)

- *Product filtering*
- *Improved UI components*
- *Error handling improvements*

Low Priority / Future Scope

- *Role-based admin system*
- *Analytics dashboard*
- *Notifications for low stock*
- *Deployment and cloud hosting*

Final Decision:
The team finalized the ***Product Inventory Management System*** *as it provides strong learning opportunities in authentication, backend API design, and frontend integration using MERN stack technologies*